# 1.3 How do computers represent text, images and sound?

You have already learnt that computers only process binary. This means that all data must be converted to binary to be processed by a computer. You have already looked at converting numbers, but text, images and sound all have to be converted to binary too, so they can be processed by a computer. How is this done?

## Converting text to binary

When a computer converts text to binary to be processed it uses a tool called a **character set**. There are two main types of character set that a computer can use, ASCII and Unicode. An ASCII character set represents characters using 8-bit binary numbers. This means that it can represent up to 256 characters (0 to 255). This is fine for certain languages, but some languages have an alphabet that requires more than 256 characters; therefore, Unicode was invented. A Unicode character set represents characters using 16-bit binary numbers. This means it can represent a much greater number of characters, approximately 65 000. This means that many more characters can be developed in different alphabets and also symbols such as emojis can also be used.

Each character has a set binary value, for example, the letter A is 01000001.

**KEY WORD**

**character set:** it contains all the characters that are in that character set and the binary value that is assigned to each character.

### ACTIVITY 1.2

Use the internet to find an example of an ASCII character set and a Unicode character set. What kind of different characters are represented? Are upper and lowercase letters the same?

Using the ASCII table, try to decode this message:

01000011011011110100110101110000011101010111010001100101011100100101001101

10001101101001010001010110111001100011011001010110100101110011011001100111

01010110111000100001

Remember, each letter will be represented as an 8-bit binary number.

Try writing a message for your friend to decode.

## Converting images to binary

An image is made up of tiny dots called **pixels**. If an image was simply created using the colours black and white, each pixel would be either black or white. The binary value 1 could be used to represent the colour black and the binary value 0 could be used to represent the colour white.

**KEY WORD**

**pixel:** a very small dot of colour that is displayed with many others to create an image.

If each pixel is converted to its binary value, a data set such as the following could be created:

0011111000100000101000000011001010011000000011010001011001110010100000100
01111100

If the computer is informed that the image that should be created using this data is 9 pixels wide and 10 pixels high, it can set each pixel to black or white and create the image (See Figure 1.2). The type of data that is used to provide information, such as the dimensions and **resolution** of an image, is called **metadata**.

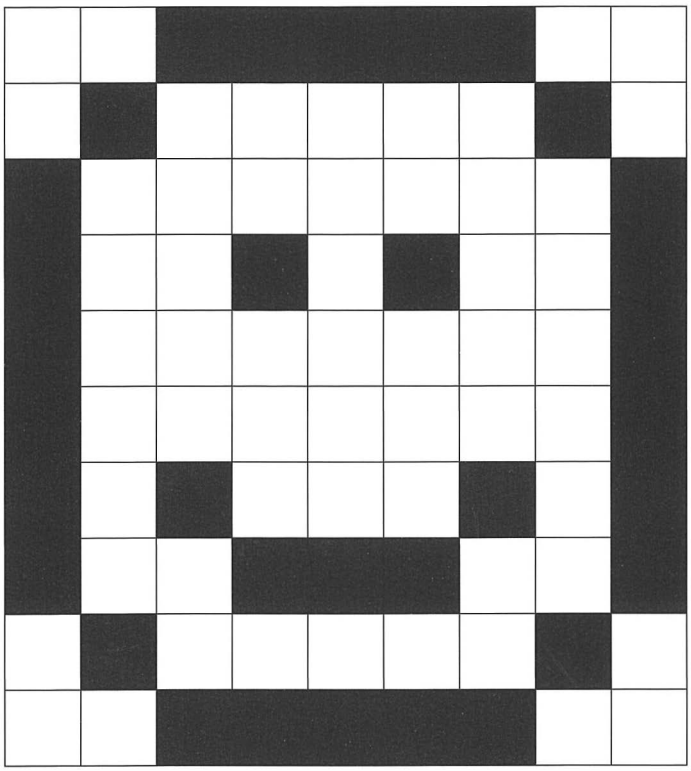


**Figure 1.2:** An image created just from black and white pixels

Most images use a lot more colours than black and white. Each colour has its own binary values. Colours are created by computer screens using the Red Green Blue (RGB) colour system. This system mixes the colours red, green and blue in different amounts to achieve each colour.

## ACTIVITY 1.3

Use the internet to find an RGB colour scale. You should notice that each colour in the scale is first presented as hexadecimal. The first two hexadecimal values are the amount of red, the second two the amount of blue and the last two the amount of green.

Practise your hexadecimal to binary conversion by converting your favourite colour to its binary value.

## KEY WORDS

**resolution:** the dimensions of an image.

**metadata:** additional data that is stored with an image that can provide information such as the dimensions of the image and the time and date the image was taken.

Most images are created using thousands of pixels. Each of those pixels needs quite a lot of data. Multiply that by the thousands of pixels that make up the image and that is a huge amount of data. This is why image files can often be so large in size.

Each image has a resolution and a **colour depth**. The resolution is the number of pixels wide and the number of pixels high that are used to create the image. The colour depth is the number of bits that are used to represent each colour. For example, each colour could be represented using 8-bit, 16-bit or 32-bit binary numbers (see Figure 1.3). The greater the number of bits, the greater the range of colours that can be represented.

KEY WORD

**colour depth:** the number of bits that are used to create each colour in an image

If the colour depth of an image is reduced, the quality of the image is often reduced.



**Figure 1.3:** 8-bit colour 16-bit colour 32-bit colour

Can you see how the quality of the image is reduced in comparison to the number of bits used?

If the resolution or the colour depth of an image is changed, this will have an effect on the size of the image file. If the resolution is increased, the image will be created using more pixels, so more data will need to be stored. If the colour depth of the image is increased, each pixel will need more data to display a greater range of colours so more data will need to be stored. Both will result in a larger file size for the image.

COMPUTER SCIENCE IN CONTEXT

People often have to sacrifice the quality of their images when displaying them online. This is because having files that are very large in size can take more time to download. This may frustrate users as it may mean they have to wait for a website to load, due to the number of large files it contains.

## Converting sound to binary

Sound is made up of sound waves. When sound is recorded, this is done at set time intervals. This process is known as **sound sampling**.

KEY WORD

**sound sampling:** a little piece of sound that is recorded at regular time intervals.

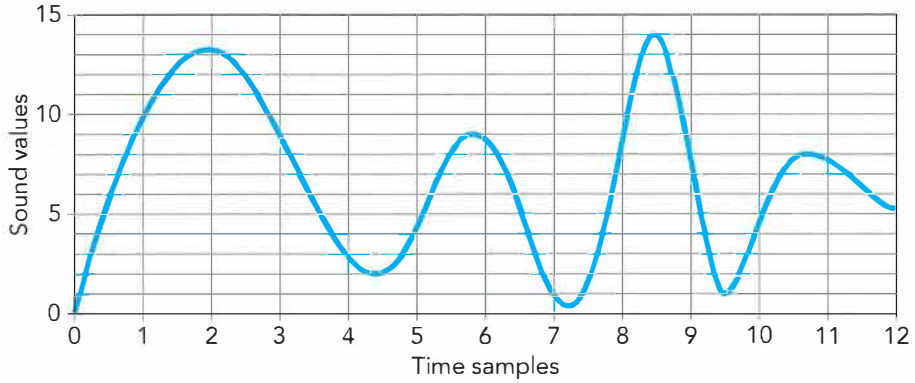



**Figure 1.4:** A sound wave

Figure 1.4 is an example of a sound wave.

If you recorded the sound at each time sample, you would have the following set of values:

| Time sample | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sound value | 9 | 13 | 9 | 3.5 | 4 | 9 | 1.5 | 9 | 8 | 5 | 8 | 5.5 |

Each sound value is converted to binary to be processed by the computer. Each sound value is then used to playback the sound, when the recording is played. If the recorded sound wave is played it would now look like Figure 1.5:

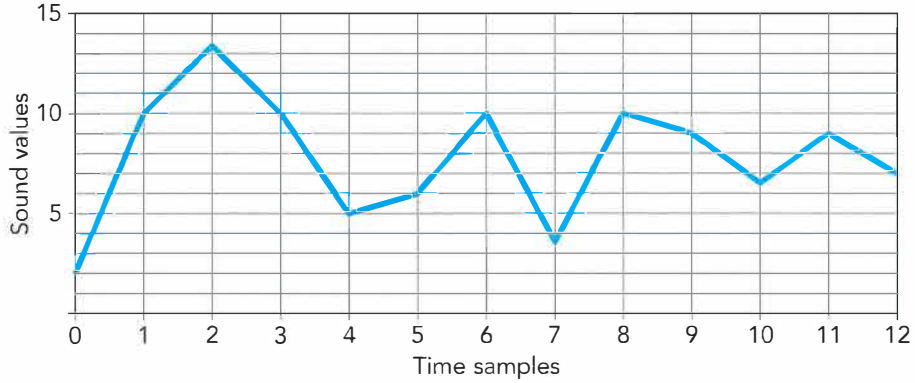



**Figure 1.5:** A sound wave created by playing back a recorded sound

You will notice that the two sound waves look different. This is because the number of samples taken is too far apart, so vital data in between time samples has been lost. This will result in a lack of accuracy in the sound that is recorded. Therefore, more samples need to be taken each second to improve the accuracy of the recorded sound. The number of samples taken in a second is called the **sample rate**. Sample rates are measured in hertz. 1 hertz is equal to 1 sample per second. A common sample rate is 44.1 khz (kilohertz), this would require 44 100 samples to be taken each second. That is a lot of data! If the sample rate is increased, the amount of data required for the recording is increased. This increases the size of the file that stores the sound.

Further data is required to represent sound, and this is the **sample resolution**. This is the number of bits that are used to represent each sample. A common sample resolution is

KEY WORDS

**sample rate:** the number of samples recorded each second.

**sample resolution:** the number of bits that are used to record each sound sample.

16-bit. The higher the sample resolution, the greater the variations in amplitude that can be stored for each sample. This means that aspects such as the loudness of the sound can be recorded more accurately. This will also increase the amount of data that needs to be stored for each sample. This increases the size of the file that stores the sound.

# 1.4 Measuring data storage

Earlier in the chapter, you learnt that all data need to be converted to digital data to be processed by a computer, and you now know how that is done for different types of data. But how is data measured when it is stored?

## COMPUTER SCIENCE IN CONTEXT

There are times when you will need to know the size of a data file, such as when you are attaching a file to an email. Most email providers have a file size limit, so you need to know the size of that file, to know if it can be attached to your email.

Data is measured using different units, which are shown in Table 1.2.

| Unit | Description |
|---|---|
| Bit | This is the smallest measurement for data |
| Nibble | There are 4 bits in a nibble |
| Byte | There are 8 bits in a byte |
| Kibibyte (KiB) | There are 1024 bytes in a Kibibyte |
| Mebibyte (MiB) | There are 1024 Kibibytes in a Mebibyte |
| Gibibyte (GiB) | There are 1024 Mebibytes in a Gibibyte |
| Tebibyte (TiB) | There are 1024 Gibibyte in a Tebibyte |
| Pebibyte (PiB) | There are 1024 Tebibytes in a Pebibyte |
| Exbibyte (EiB) | There are 1024 Pebibytes in an Exbibyte |

**Table 1.2:** Data measuring units

Although most computers will calculate the size of a file for you, you need to be able to show that you can do this manually for the exam. This is just for an image file and a sound file.

## SKILLS FOCUS 1.3

### CALCULATING THE SIZE OF A FILE

There are several pieces of information that you need to know to calculate the size of an image file, these are:

- the number of images in the file
- the resolution of the image
- the colour depth of the image.